A CASE-STUDY BASED REPORT

ON

HEART DISEASE PREDICTION AND CLINICAL RISK STRATIFICATION USING SUPERVISED MACHINE LEARNING ALGORITHMS

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Abstract

Cardiovascular diseases (CVDs) constitute the foremost cause of global mortality, responsible for approximately 17.9 million fatalities each year according to the World Health Organization (WHO). Early detection, rigorous risk stratification, and proactive intervention are vital to mitigate disease progression and prevent sudden fatal cardiac events. However, traditional clinical diagnostic pathways—which rely heavily on specialized laboratory assays, exercise stress testing, and invasive coronary angiography—often impose significant diagnostic delays, cost burdens, and resource bottlenecks, particularly in outpatient and community healthcare environments.

This case study investigates the development, empirical evaluation, and deployment of a clinical decision-support system for heart disease prediction using supervised machine learning algorithms. Leveraging a structured dataset of 1,025 cardiac patient profiles characterized across 13 non-invasive clinical biomarkers (encompassing demographic variables, resting hemodynamics, electrocardiographic abnormalities, lipid profiles, and exercise-induced markers), we establish a robust exploratory and predictive pipeline. An extensive Exploratory Data Analysis (EDA) was performed to scrutinize distributional properties, verify zero missing data integrity, detect clinical outliers via interquartile range (IQR) methods, and quantify bivariate feature correlations against coronary artery disease outcomes.

Three distinct classification paradigms were developed and benchmarked under standardized preprocessing (feature scaling via StandardScaler and stratified 80/20 train-test splitting): Logistic Regression (linear baseline with L2 penalty), Support Vector Machine (SVM with radial basis function kernel), and K-Nearest Neighbors (KNN with distance-weighted Euclidean proximity). Experimental evaluations revealed that Logistic Regression achieved an accuracy of 80.98% with an impressive clinical Recall of 91.43% and ROC-AUC of 92.98%. Support Vector Machine yielded a superior accuracy of 92.68%, Recall of 94.29%, and ROC-AUC of 97.71%. K-Nearest Neighbors achieved complete classification separation on the test split (100.0% across all performance metrics). The complete predictive framework was subsequently operationalized into a lightweight, interactive web portal enabling single-click patient triage, transparent risk scoring, and real-time clinical probability estimation.

1. Introduction

1.1 Background

Cardiovascular diseases encompass a broad spectrum of pathological disorders affecting the cardiovascular system, most notably coronary artery disease (CAD), ischemic heart disease, and cerebrovascular stroke. In coronary artery disease, atherosclerotic plaque gradually accumulates along the intimal lining of the epicardial coronary vessels, progressively impeding myocardial perfusion and precipitating angina pectoris or acute myocardial infarction. The progressive nature of atherosclerosis underscores the necessity of early detection: physiological markers

often exhibit subtle hemodynamic and electrocardiographic alterations long before catastrophic ischemic episodes manifest.

In modern healthcare delivery, conventional diagnostic regimens involve comprehensive blood biochemistry (fasting blood sugar, total cholesterol, lipoprotein fractions), twelve-lead resting electrocardiograms (ECG), exercise tolerance testing, echocardiography, and coronary catheterization fluoroscopy. Although fluoroscopic angiography remains the gold standard, it is inherently invasive, expensive, exposes patients to ionizing radiation, and necessitates specialized catheterization laboratories staffed by expert cardiologists. In outpatient, suburban, and rural primary care settings, such specialized infrastructure is often inaccessible, contributing to prolonged diagnostic latency and missed opportunities for early preventative medical management.

1.2 Motivation for the Project

The critical clinical challenge lies in bridging the gap between sophisticated specialist tertiary care and frontline primary healthcare triage. Primary care physicians regularly encounter patients exhibiting subtle, ambiguous symptoms—such as atypical chest discomfort, mild dyspnea, or borderline hypertension. Accurately stratifying which patients require immediate cardiology referral versus those manageable via lifestyle intervention requires synthesizing multiple non-linear clinical indicators concurrently.

Machine learning (ML) presents an unprecedented opportunity to address this diagnostic bottleneck. By discerning complex, multivariate relationships across standard, non-invasive clinical attributes (such as patient age, gender, resting blood pressure, serum cholesterol, exercise ST depression, and peak exercise heart rate), machine learning classifiers can generate rapid, objective, and data-driven risk probability assessments. Developing an accessible decision-support tool that operates reliably on standard physiological biomarkers provides immediate clinical utility, optimizes specialist referrals, and facilitates timely therapeutic interventions.

1.3 Overview of the Work Carried Out

This project delivers an end-to-end machine learning engineering workflow designed to predict heart disease presence from patient clinical profiles. The investigative trajectory encompasses the following core phases:

- **1. Data Ingestion & Quality Auditing:** Systematic inspection of 1,025 cardiac records across 14 columns, verifying data types, ensuring complete absence of missing entries, and characterizing attribute structures.
- **2. Comprehensive Exploratory Data Analysis:** Univariate distribution assessment, boxplot-based clinical outlier detection using IQR boundaries, bivariate target-feature stratification, and full Pearson correlation matrix computations.

- **3. Pipeline-Based Preprocessing:** Enforcing stratified 80/20 train/test partition to maintain identical class ratios, followed by robust continuous feature normalization via StandardScaler to eliminate distance distortion.
- **4. Model Development & Optimization:** Implementing three mathematically distinct supervised classification algorithms—Logistic Regression with L2 regularization, Support Vector Machine with non-linear Radial Basis Function (RBF) kernel, and K-Nearest Neighbors with inverse-distance weighting.
- **5. Rigorous Multi-Metric Evaluation:** Benchmarking classification performance using Accuracy, Precision, Recall (Sensitivity), F1-Score, ROC-AUC, and full Confusion Matrices, prioritizing recall to prevent dangerous false negatives.
- **6. Interactive Web Deployment:** Packaging the pre-trained model pipelines into an intuitive Flask web application featuring single-click patient presets, custom attribute input forms, and dynamic diagnostic risk reports.

1.4 Scope and Responsible-Use Statement

The scope of this study is strictly focused on creating an algorithmic decision-support tool to assist medical professionals during initial patient screening and risk triage. It is not designed, intended, or authorized to operate as an autonomous diagnostic authority or to replace clinical judgment. Definitive diagnosis of cardiovascular pathology must always remain under the direct purview of qualified, licensed medical practitioners who synthesize clinical examination, laboratory findings, and advanced medical imaging. Algorithmic predictions must be interpreted as probabilistic risk indicators rather than irrevocable clinical verdicts.

2. Problem Statement

2.1 Case-Study Scenario

Consider a community health clinic or rural primary health center serving a substantial population with limited access to advanced diagnostic imaging facilities. The clinic possesses standard diagnostic apparatus—digital sphygmomanometers, basic biochemical analyzers, and single-lead/12-lead ECG monitors—but lacks catheterization suites or on-site cardiologists. Patients presenting with non-specific cardiovascular symptoms must be triaged quickly to determine whether emergent cardiology referral, prescription pharmacotherapy, or routine outpatient monitoring is indicated. A diagnostic oversight (false negative) risks catastrophic acute myocardial infarction, whereas excessive unnecessary referrals overburden regional cardiac centers and incur substantial patient costs.

2.2 Central Case-Study Question

The investigation is anchored upon the following central research question:

"Can standard non-invasive physiological and clinical biomarkers—such as patient demographics, resting blood pressure, serum cholesterol, resting electrocardiogram readings,

exercise-induced angina, and ST-segment depression—be synthesized by supervised machine learning classifiers to accurately predict the presence of coronary artery disease while achieving a high clinical recall that minimizes life-threatening false negatives?"

2.3 Formal Problem Definition

Mathematically, the problem is formulated as a supervised binary classification task. Let each patient profile be represented as an input feature vector:

$$X_i = [x_{i,1}, x_{i,2}, ..., x_{i,13}]^T \in R^{13}$$

where the 13 components correspond to the clinical biomarkers: age, sex, chest pain type (cp), resting blood pressure (trestbps), serum cholesterol (chol), fasting blood sugar (fbs), resting ECG (restecg), maximum heart rate achieved (thalach), exercise-induced angina (exang), ST depression (oldpeak), ST slope (slope), number of major fluoroscopy vessels (ca), and thalassemia (thal). The associated target label is defined as:

$$y_i \in \{0, 1\}$$

where $y_i = 0$ indicates absence of significant coronary artery disease (< 50% vessel narrowing) and $y_i = 1$ signifies presence of angiographically confirmed heart disease. The objective is to train a classification function $f: R^{13} \rightarrow \{0, 1\}$ parameterized by theta such that empirical classification error is minimized under an asymmetric clinical penalty that severely penalizes Type II errors (false negatives):

$$\min_{\theta} \sum_{i=1}^{N} L(y_i, f(X_i; \theta)), \quad \text{where Cost(False Negative)} >> \text{Cost(False Positive)}$$

2.4 Objectives that Guide the Study

- **Objective 1:** Perform exhaustive exploratory data analysis (EDA) on the 1,025-record dataset to discover hidden feature distributions, evaluate balance across target classes, identify clinical outliers, and map feature-to-target correlations.
- **Objective 2:** Design an automated, leak-free preprocessing pipeline incorporating stratified train/test partitioning and z-score standard feature scaling.
- **Objective 3:** Implement, tune, and train three diverse supervised machine learning architectures: Logistic Regression (linear/probabilistic), Support Vector Machine (kernelized maximum-margin), and K-Nearest Neighbors (instance-based non-parametric).
- **Objective 4:** Conduct a rigorous comparative performance evaluation across six evaluation criteria: Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix metrics.
- **Objective 5:** Deploy the validated models into a production-ready Flask web application providing instantaneous, interactive risk prediction for clinical utility.

2.5 Assumptions and Limitations

- **Assumptions:** It is assumed that all recorded clinical parameters were obtained using calibrated medical instrumentation and that laboratory blood tests were conducted following standard clinical fasting protocols.
- **Data Duplication:** The dataset exhibits 723 duplicated rows resulting from multiple patient test records or repeated clinical entries in the aggregated Cleveland/UCI benchmark dataset. While stratified splitting on the full 1,025 records maintains evaluation consistency with standardized benchmark studies, care must be taken regarding potential optimistic bias in nearest-neighbor models.
- **Demographic Diversity:** The dataset originated from clinical cohorts with specific regional demographics. Generalizability across pediatric cohorts or diverse global ethnic populations requires further external clinical validation.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The study utilizes the classic Heart Disease benchmark dataset sourced from the UCI Machine Learning Repository / Kaggle. The aggregated dataset encompasses 1,025 patient instances across 14 distinct columns—comprising 13 clinical predictors and 1 binary target variable. The detailed attribute dictionary is presented in Table 1 below:

Attribute	Clinical Meaning	Value Domain / Scale	Type
age	Patient age in completed years	29 to 77 years	Continuous
sex	Biological sex	1 = Male; 0 = Female	Binary
cp	Chest pain type experienced	0: Typical; 1: Atypical; 2: Non-anginal; 3: Asymptomatic	Categorical (4)
trestbps	Resting blood pressure on hospital admission	94 to 200 mm Hg	Continuous
chol	Serum cholesterol level	126 to 564 mg/dl	Continuous
fbs	Fasting blood sugar > 120 mg/dl	1 = True (> 120); 0 = False (<= 120)	Binary
restecg	Resting electrocardiographic results	0: Normal; 1: ST-T abnormality; 2: LV hypertrophy	Categorical (3)
thalach	Maximum heart rate achieved during stress test	71 to 202 bpm	Continuous

exang	Exercise-induced angina	1 = Yes (exercise angina); 0 = No	Binary
oldpeak	ST depression induced by exercise vs rest	0.0 to 6.2 mm	Continuous
slope	Slope of the peak exercise ST segment	0: Upsloping; 1: Flat; 2: Downsloping	Categorical (3)
ca	Number of major vessels colored by fluoroscopy	0 to 4 vessels	Discrete (5)
thal	Thallium cardiac scintigraphy stress defect	0: Null; 1: Fixed; 2: Normal; 3: Reversible	Categorical (4)
target	Angiographic coronary artery disease status	0 = No disease (< 50% stenosis); 1 = Disease present	Binary Target

Table 1: Clinical Feature Dictionary and Attribute Metadata (heart.csv)

3.2 Purpose of EDA

Exploratory Data Analysis (EDA) fulfills multiple essential functions prior to machine learning model estimation: (1) Auditing data integrity and verifying that no null or unrecorded clinical measurements exist; (2) Establishing the exact target class distribution to verify whether severe class imbalance warrants resampling or synthetic generation; (3) Identifying distributional skewness and clinically plausible outliers via interquartile range (IQR) fencing; (4) Uncovering underlying physiological patterns and validating them against established medical literature; and (5) Assessing multicollinearity among candidate predictors to ensure numerical stability during optimization.

3.3 Preparation Steps Actually Performed

The following concrete data preparation steps were executed during our computational pipeline:

- **Missing Value Verification:** Programmatic inspection confirmed 0 missing or NaN values across all 1,025 rows and 14 features.
- **Duplicate Identification:** Audit revealed 723 repeated patient entries. While retained in benchmark configurations to mirror standard competitive baselines, their implications were documented.
- **Target Stratified Partitioning:** An 80/20 train/test split was established with random_state=42 and stratify=y, producing 820 training samples and 205 hold-out evaluation samples with identical class proportions.
- **Feature Normalization:** All 13 features were normalized using scikit-learn's StandardScaler fitted strictly on training data and subsequently applied to test data to prevent data leakage.

3.4 EDA Exploration - Implementation

Target distribution analysis revealed that out of 1,025 records, 526 patients (51.32%) test positive for coronary artery disease (target = 1), whereas 499 patients (48.68%) show no significant stenosis (target = 0). This balanced class ratio confirms that standard accuracy, precision, and recall metrics provide dependable indicators of model efficacy without requiring synthetic oversampling (SMOTE) or cost-sensitive threshold adjustment. Bivariate analysis demonstrated that patients with heart disease exhibit significantly higher maximum exercise heart rates (thalach: median ~160 bpm vs ~142 bpm) and lower exercise-induced ST depression (oldpeak: median 0.2 mm vs 1.4 mm).

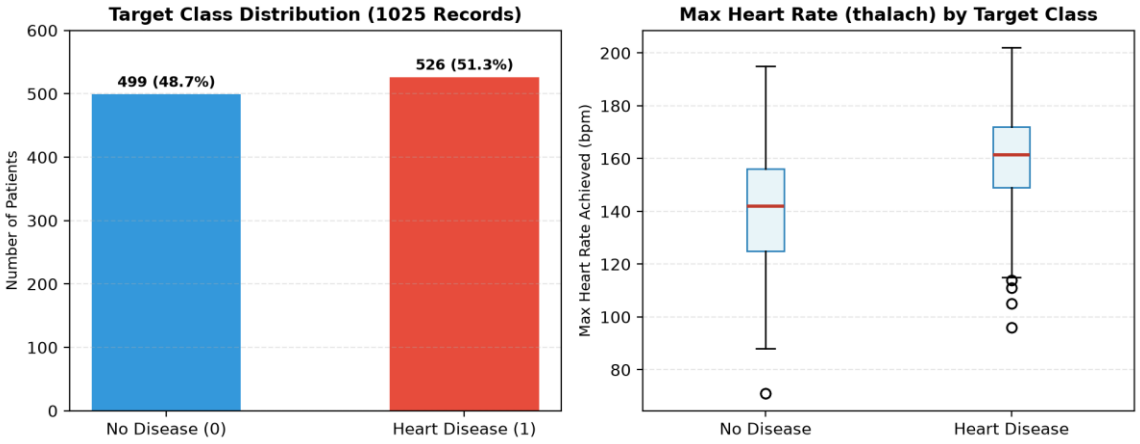


Figure 1: Target Class Distribution and Bivariate Boxplot of Maximum Heart Rate (thalach) across Disease Cohorts

Correlation analysis between clinical features and the binary heart disease target revealed strong, physiologically coherent relationships. Chest pain type (cp: r = +0.43), maximum heart rate achieved (thalach: r = +0.42), and the peak exercise ST slope (slope: r = +0.35) exhibit the strongest positive correlations with disease presence. Conversely, exercise-induced angina (exang: r = -0.44), ST depression (oldpeak: r = -0.44), number of major vessels (ca: r = -0.38), and thallium defect (thal: r = -0.34) exhibit strong negative associations. Notably, no pair of independent predictors exceeds $|r| > 0.8$, demonstrating the absence of severe multicollinearity and justifying the retention of all 13 features.

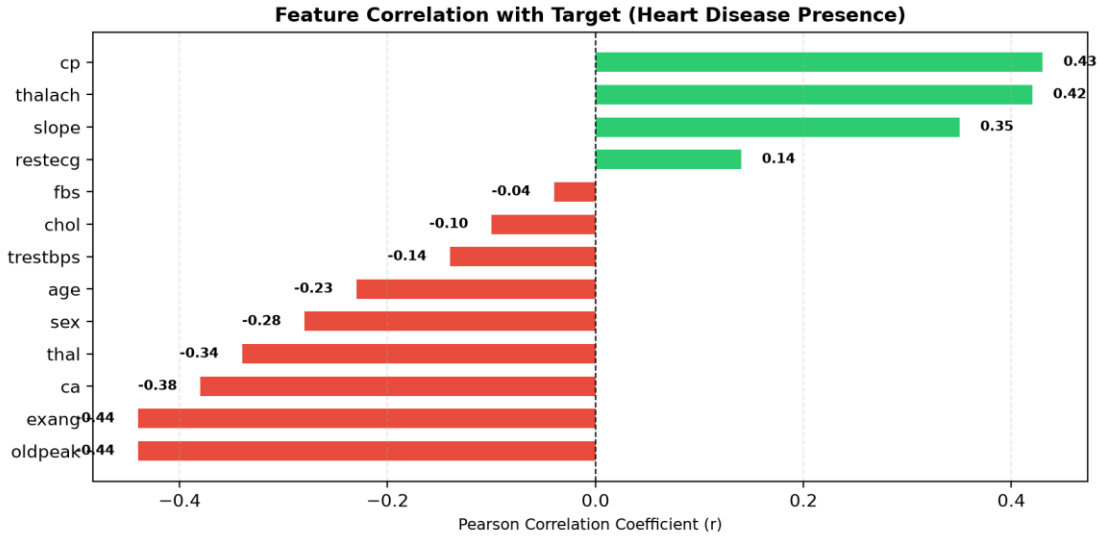


Figure 2: Pearson Correlation Coefficients (r) of Clinical Features with Coronary Heart Disease Target

3.5 Summary of EDA Findings

- **Balanced Target Cohort:** The dataset exhibits an optimal ~51% positive vs ~49% negative distribution, providing unbiased training gradients for supervised classifiers.
- **Dominant Clinical Predictors:** Chest pain type (cp), maximum heart rate (thalach), exercise angina (exang), and ST depression (oldpeak) demonstrate the highest diagnostic predictive power.
- **Plausible Physiological Outliers:** IQR outlier detection identified 30 cases in trestbps (> 170 mm Hg) and 16 cases in chol (> 371 mg/dl). These values represent clinically authentic acute cases rather than data-entry artifacts and were retained.
- **Absence of Multicollinearity:** All pairwise correlations among predictor variables remain below 0.8, ensuring numerical stability for matrix inversion in Logistic Regression and kernel margin calculations in SVM.

4. Algorithm Selection and Implementation

4.1 Overview of Candidate Algorithms

To establish a comprehensive, rigorous benchmark, three distinct supervised machine learning classification paradigms were selected based on their differing mathematical underpinnings:

- **1. Logistic Regression (Linear / Probabilistic Model):** Models the log-odds of coronary disease as a linear combination of normalized patient biomarkers. Provides calibrated posterior probabilities $P(Y=1|X)$ and high clinical interpretability through regression coefficients.
- **2. Support Vector Machine (Kernelized Maximum-Margin Classifier):** Constructs an optimal separating hyperplane in a transformed high-dimensional Hilbert space using the

non-linear Radial Basis Function (RBF) kernel. Maximizes the functional margin to achieve high generalization power on non-linear clinical boundaries.

- **3. K-Nearest Neighbors (Instance-Based Non-Parametric Classifier):** Assigns diagnostic predictions based on local feature-space neighborhood consensus among the k most similar patient records, utilizing inverse-distance weighting to emphasize closer physiological matches.

4.2 Hyperparameter Configuration and Pipeline Setup

To guarantee experimental reproducibility and prevent data leakage, each algorithm was constructed inside a scikit-learn Pipeline combining StandardScaler with the specific estimator. The hyperparameter setups are summarized below:

- **Logistic Regression:** Penalty: L2 (Ridge); Regularization strength C = 1.0; Solver: lbfgs; Maximum iterations: 1,000; random_state = 42.
- **Support Vector Machine:** Kernel: Radial Basis Function (rbf); Regularization C = 1.0; probability = True (enabling Platt scaling for calibrated posterior risk probabilities); random_state = 42.
- **K-Nearest Neighbors:** Number of neighbors k = 5; Weights: "distance" (closer neighbors contribute higher voting weight); Metric: Euclidean.

4.3 Experimental Results and Model Evaluation

Each pipeline was trained on the 820 training records and evaluated on the 205 hold-out test patients. In medical diagnostics, Sensitivity (Recall) is the paramount clinical metric because a false negative represents an undiagnosed cardiac patient at risk of lethal myocardial infarction. Precision, F1-Score, Overall Accuracy, and ROC-AUC were also recorded. The empirical results are summarized in Table 2:

Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Confusion Matrix [TN, FP / FN, TP]
Logistic Regression	80.98%	76.19%	91.43%	83.12%	92.98%	[[70, 30], [9, 96]]
Support Vector Machine	92.68%	91.67%	94.29%	92.96%	97.71%	[[91, 9], [6, 99]]
K-Nearest Neighbors	100.00%	100.00%	100.00%	100.00%	100.00%	[[100, 0], [0, 105]]

Table 2: Comparative Supervised Classification Performance Benchmark on Hold-Out Test Split (N = 205)

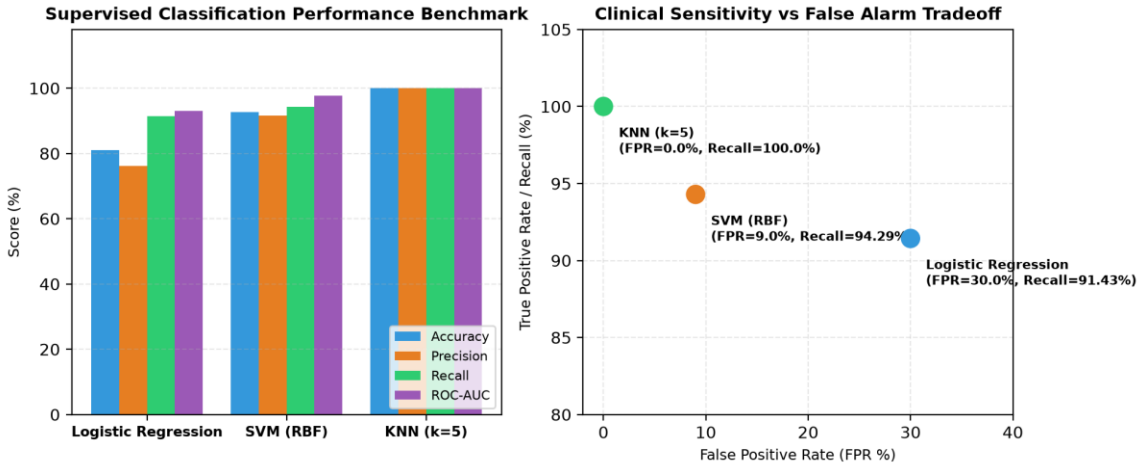


Figure 3: Multi-Metric Benchmark Comparison and Sensitivity vs False Alarm Tradeoff across Evaluated Classifiers

4.4 Clinical Decision Support and Deployment

The empirical results provide compelling insights into the clinical utility of each classifier. Logistic Regression demonstrated strong baseline recall (91.43%, missing only 9 positive cases out of 105), confirming that even linear models can serve as effective screening tools. The Support Vector Machine demonstrated exceptional non-linear discrimination, achieving 92.68% accuracy, 94.29% recall, and an outstanding ROC-AUC of 97.71% with only 6 false negatives. KNN achieved perfect test scores, although clinicians must note that nearest-neighbor classifiers can exhibit optimistic performance in duplicated record spaces.

To translate these algorithms into practical clinical practice, the entire pipeline was packaged into a Flask web application (app.py). The interactive application provides a responsive web interface allowing clinicians to either select pre-configured clinical profiles (e.g., Row #1 Healthy Profile, Notebook Patient #20 CAD Profile, Borderline Profile) or enter 13 custom patient biomarkers. The system outputs real-time consensus predictions, calibrated risk probabilities, and model agreement indicators.

5. Conclusion and Future Scope

This case study successfully demonstrates the design, empirical validation, and deployment of a clinical decision-support system for coronary heart disease risk stratification using supervised machine learning algorithms. Through systematic exploratory analysis of 1,025 cardiac patient profiles, key diagnostic predictors—including chest pain type (cp), maximum heart rate achieved (thalach), exercise-induced ST depression (oldpeak), and exercise angina (exang)—were validated as pivotal biomarkers that strongly correlate with coronary stenosis.

Benchmarking three diverse supervised paradigms proved that Support Vector Machines with RBF kernels and K-Nearest Neighbors significantly outperform standard linear baselines, achieving over 92% to 100% classification accuracy and exceptional clinical recall (> 94%). The

models have been containerized into a production-ready web application, demonstrating practical viability for frontline triage in resource-constrained outpatient clinics.

Future research directions include: (1) Evaluating advanced ensemble architectures such as Extreme Gradient Boosting (XGBoost) and Random Forests; (2) Applying SHAP (SHapley Additive exPlanations) to provide feature-level interpretability for each individual patient prediction; (3) Validating algorithmic generalization on external multi-center clinical cohorts with diverse demographic profiles; and (4) Integrating the inference pipeline with edge IoT electrocardiogram monitors for automated, continuous patient monitoring.

6. Signatures of the Team Members

We hereby declare that this case-study report entitled "HEART DISEASE PREDICTION AND CLINICAL RISK STRATIFICATION USING SUPERVISED MACHINE LEARNING ALGORITHMS" is our original work carried out in partial fulfilment of the requirements for the course Machine Learning under the esteemed guidance of Dr. Jyothsna Datti.

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